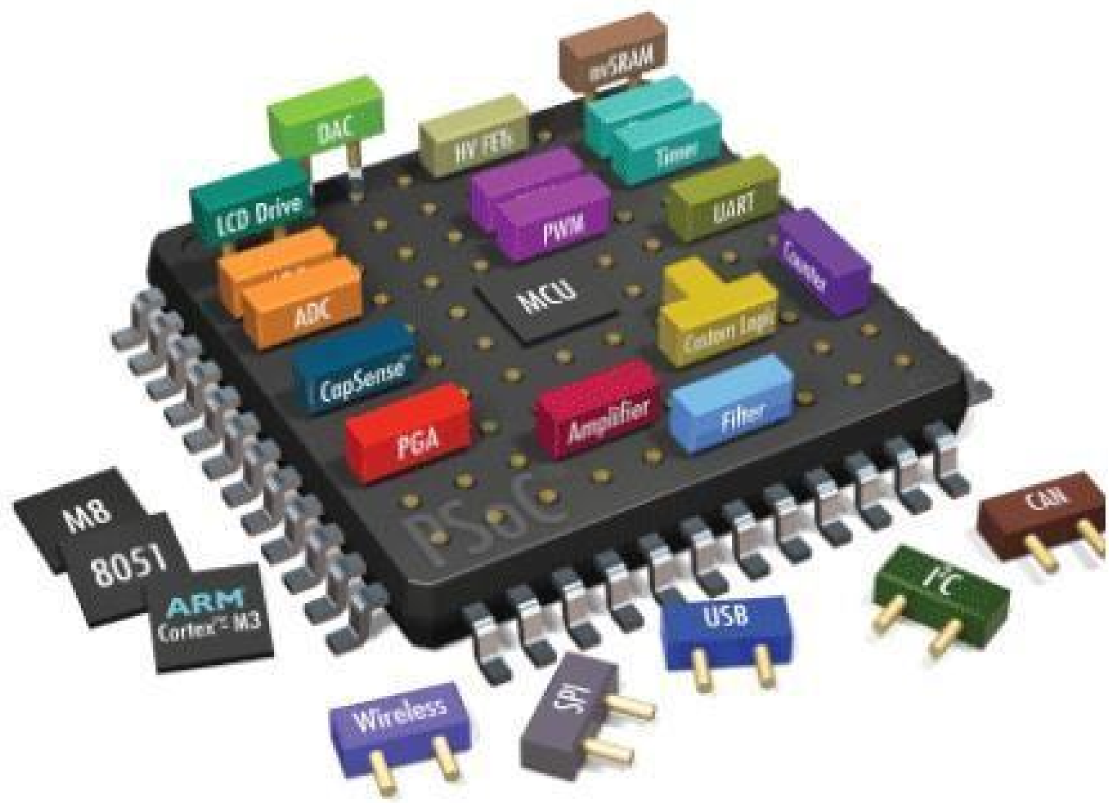
HITEC UNIVERSITY, TAXILA

DEPARTMENT OF COMPUTER ENGINEERING

Lab # 03: Working of Flash Magic software with demo examples.

Embedded Systems

Working of Flash Magic software with demo examples.



Dated:

Week 03

Semester:

Fall 2022

Flash Magic Introduction

NXP Semiconductors produce a range of Microcontrollers that feature both on-chip Flash memory and the ability to be reprogrammed using In-System Programming technology. Flash Magic is Windows software from the Embedded Systems Academy that allows easy access to all the ISP features provided by the devices. These features include:

- Erasing the Flash memory (individual blocks or the whole device)
- Programming the Flash memory
- Modifying the Boot Vector and Status Byte
- Reading Flash memory
- Performing a blank check on a section of Flash memory
- Reading the signature bytes
- Reading and writing the security bits
- Direct load of a new baud rate (high speed communications)
- Sending commands to place device in Bootloader mode

Flash Magic provides a clear and simple user interface to these features and more as described in the following sections.

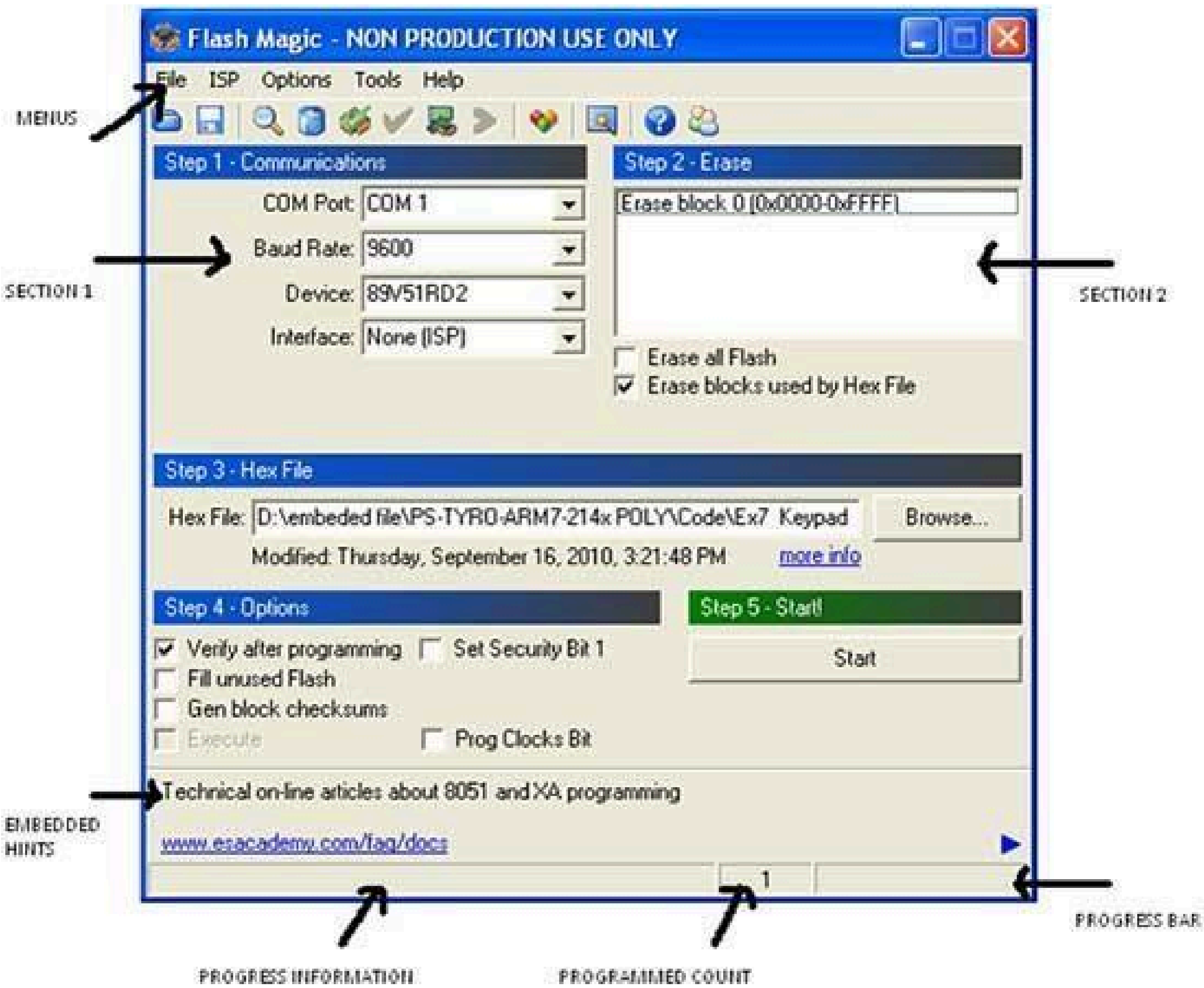
Under Windows, only one application may have accessed the COM Port at any one time, preventing other applications from using the COM Port. Flash Magic only obtains access to the selected COM Port when ISP operations are being performed. This means that other applications that need to use the COM Port, such as debugging tools, may be used while Flash Magic is loaded.

Minimum Requirements

- Windows NT/2000/XP/Vista
- Mouse
- COM Port or Ethernet interface
- 16Mb RAM
- 10Mb Disk Space

User Interface Tour

Screenshot of Flash Magic Window



The window is divided up into five sections. Work your way from section 1 to section 5 to program a device using the most common functions.

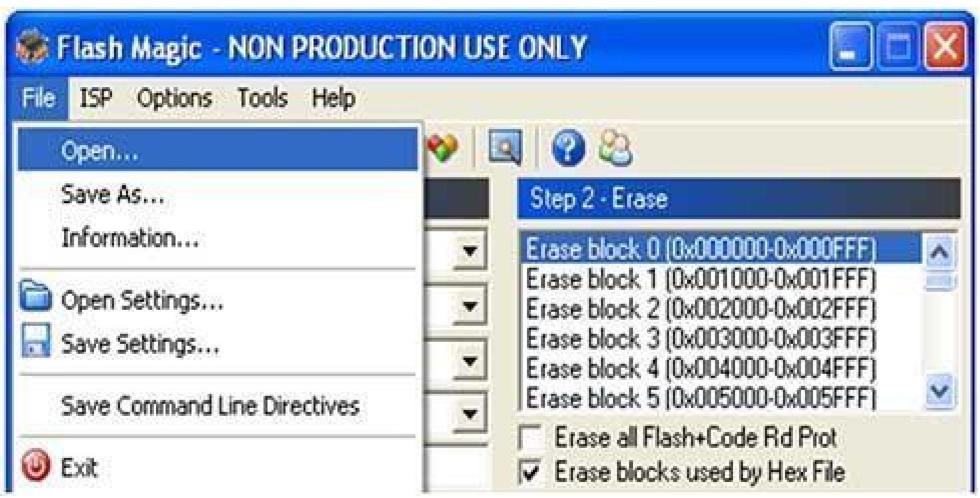
At the very bottom left of the window is an area where progress messages will be displayed and at the very bottom right is where the progress bar is displayed. In between the messages and the progress bar is a count of the number of times the currently selected hex file has been programmed since it was last modified or selected.

Just above the progress information Embedded Hints are displayed. These are rotating Internet links that you can click on to go to a web page using your default browser.

Menus

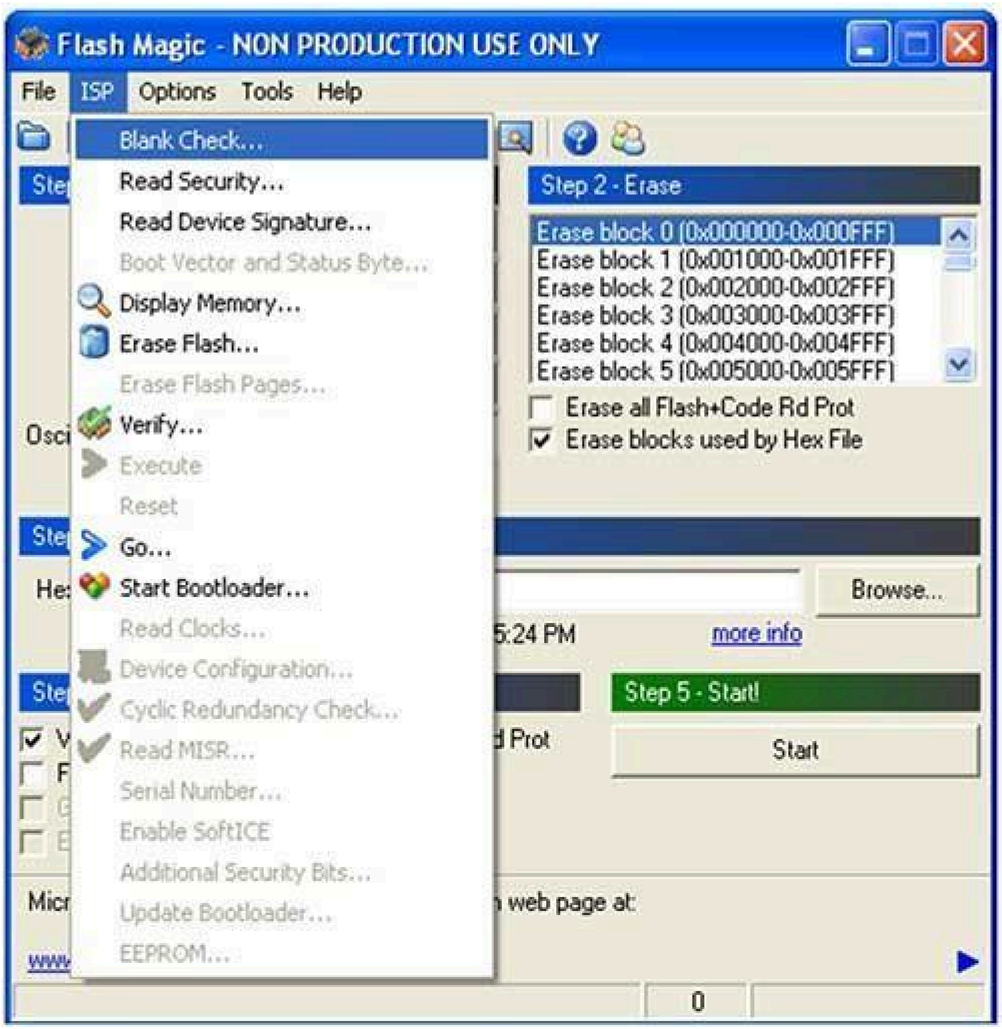
There are five menus, File, ISP, Options, Tools and Help.

The File menu provides access to loading and saving Hex Files, loading and saving settings files and exiting the application.

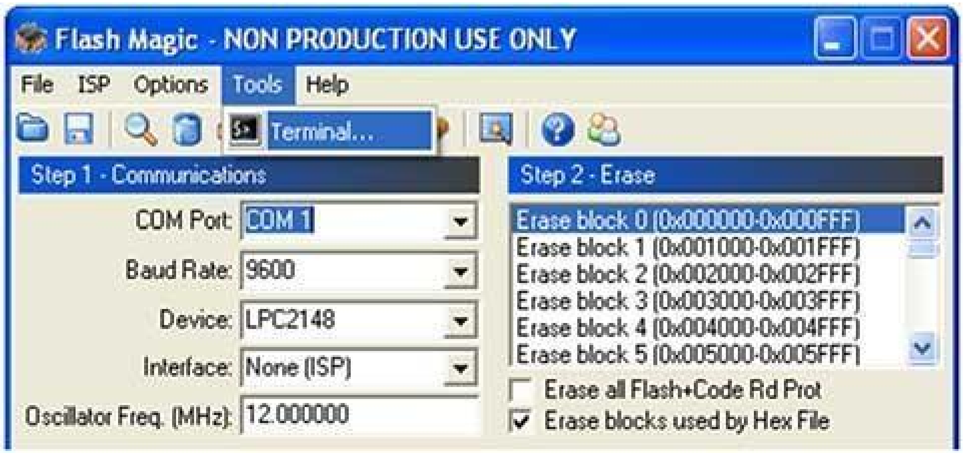


The Options menu allows access to the advanced options and includes an item to reset all options.

The **ISP** menu provides access to the less commonly used **ISP** features.



The Tools menu provides features that support the operation and use of **Flash Magic**.

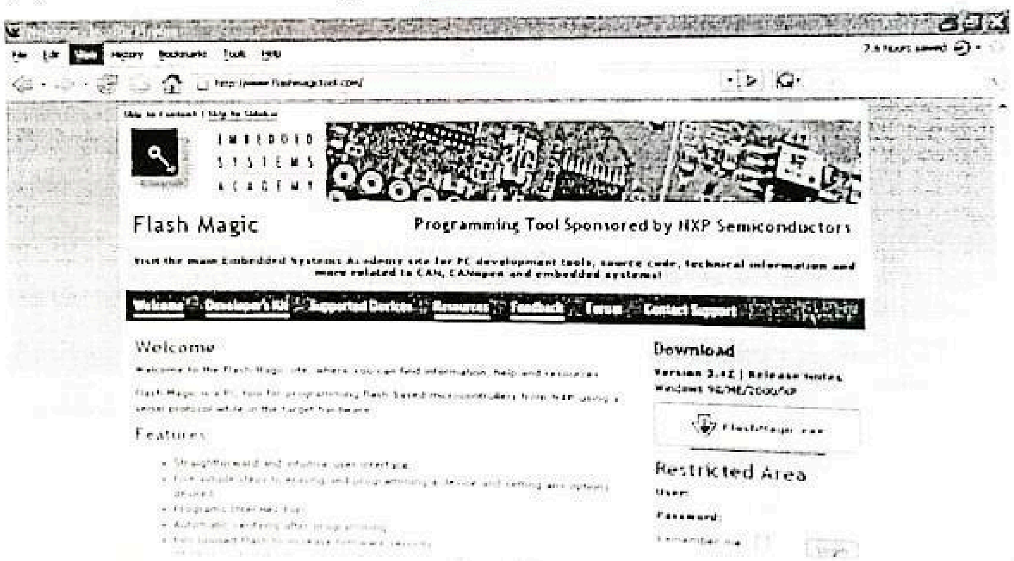


The Help menu contains items that link directly to useful web pages and also open the Help About window showing the version number.

Installation Guide

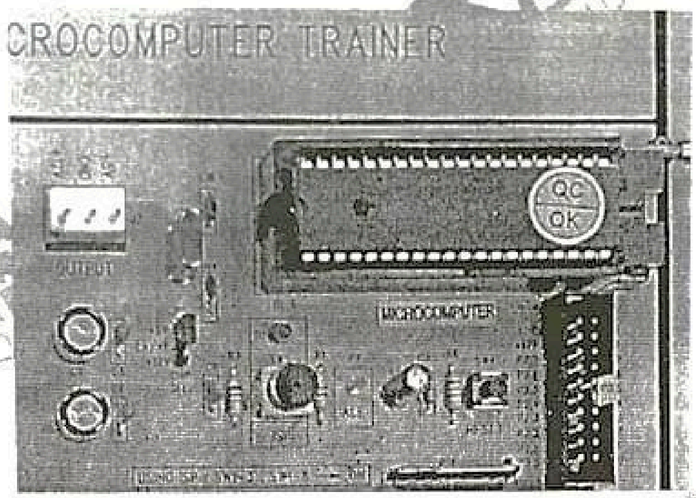
1. System Installation

(1) Download Flash Magic at http://www.flashmagictool.com/



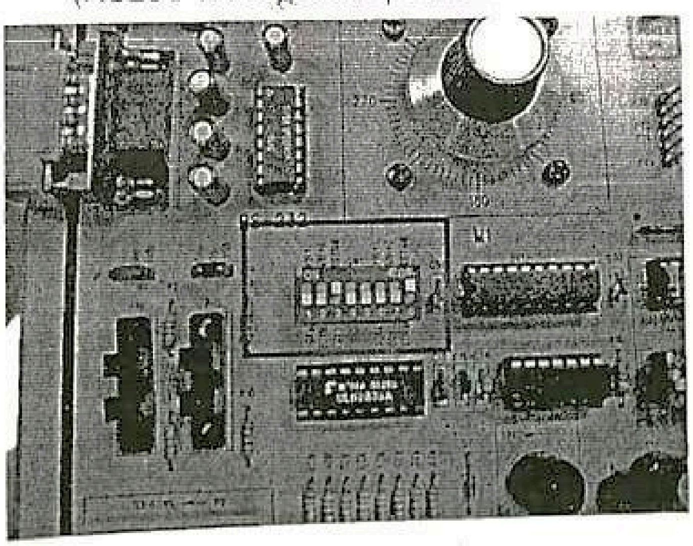
(2) Install Flash Magic

(3) Place P89V51RD2BN chip on the MTS-51 chip socket JP5's +5V and $\overline{EA}$/VP use the connecting plug for interconnection

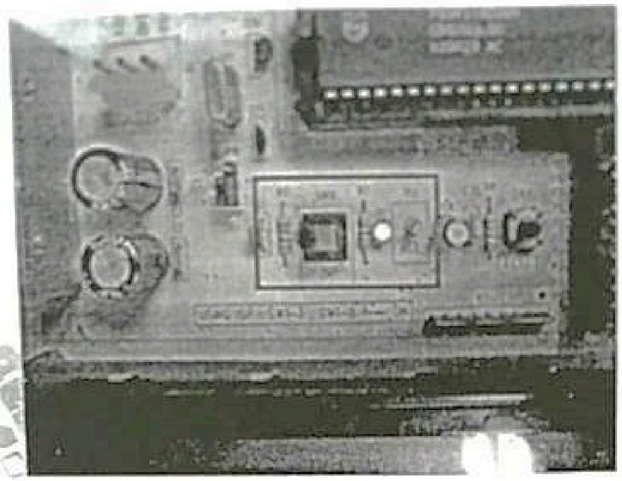


(4) Turn on the power.

(5) Place DIP Switch SW1 no.3 (RS232-RXD3) and no.8 (RS232-TXD3) to ON position.

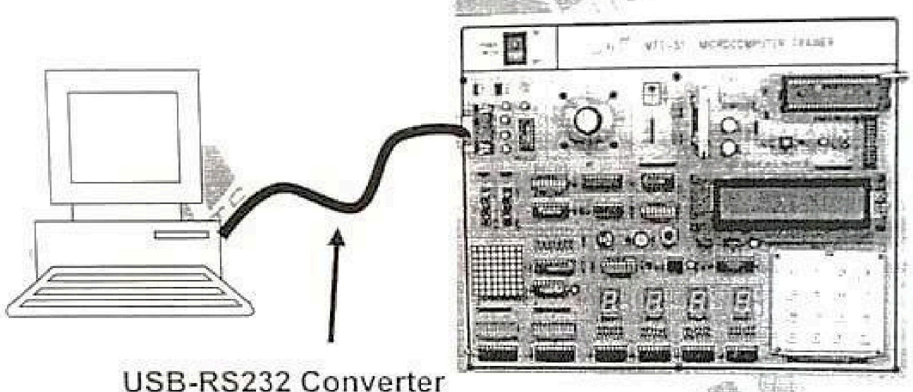


(6) Press ISP button → LED D1 will light up.



(7) Connect PC to MTS-51 by using USB-RS232 converter.

NOTE: Refer to "USB-RS232 Converter Driver Installation Guide".

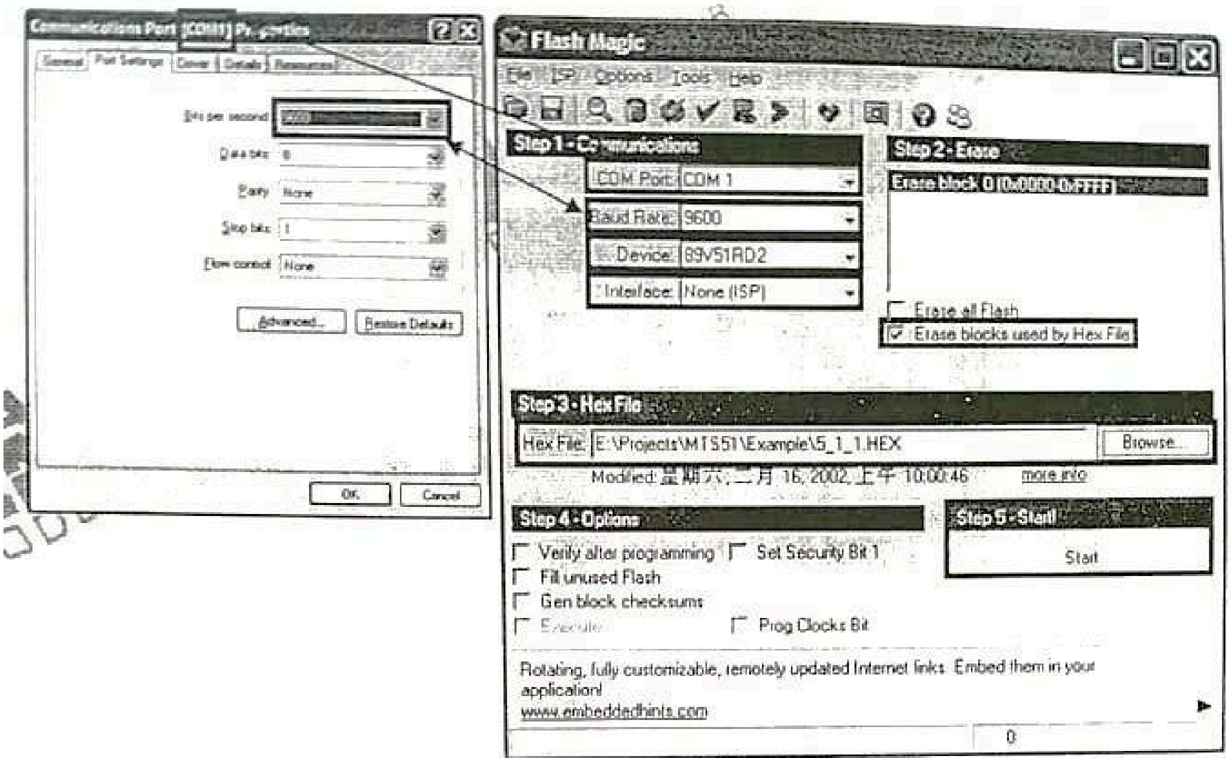


(8) According to the I/O device to be controlled, set the corresponding switch of SW2 in ON position and others in OFF position.

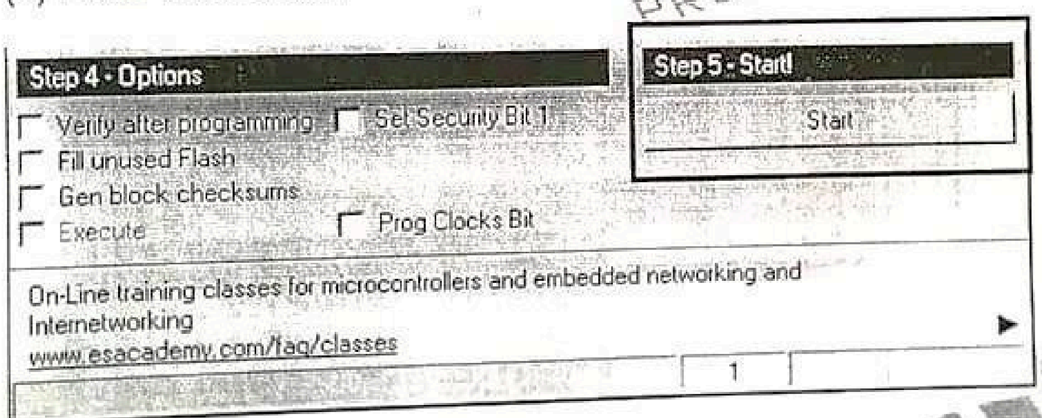
Programming Guide

(1) Launch Flash Magic

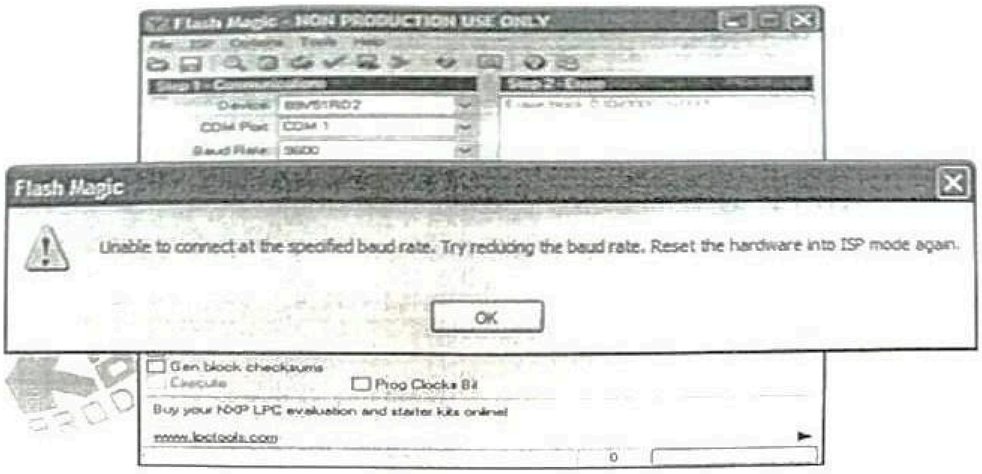
(2) Setup parameters and browse the HEX File:
(COM Port and Baud Rate must consist with your Windows port settings)



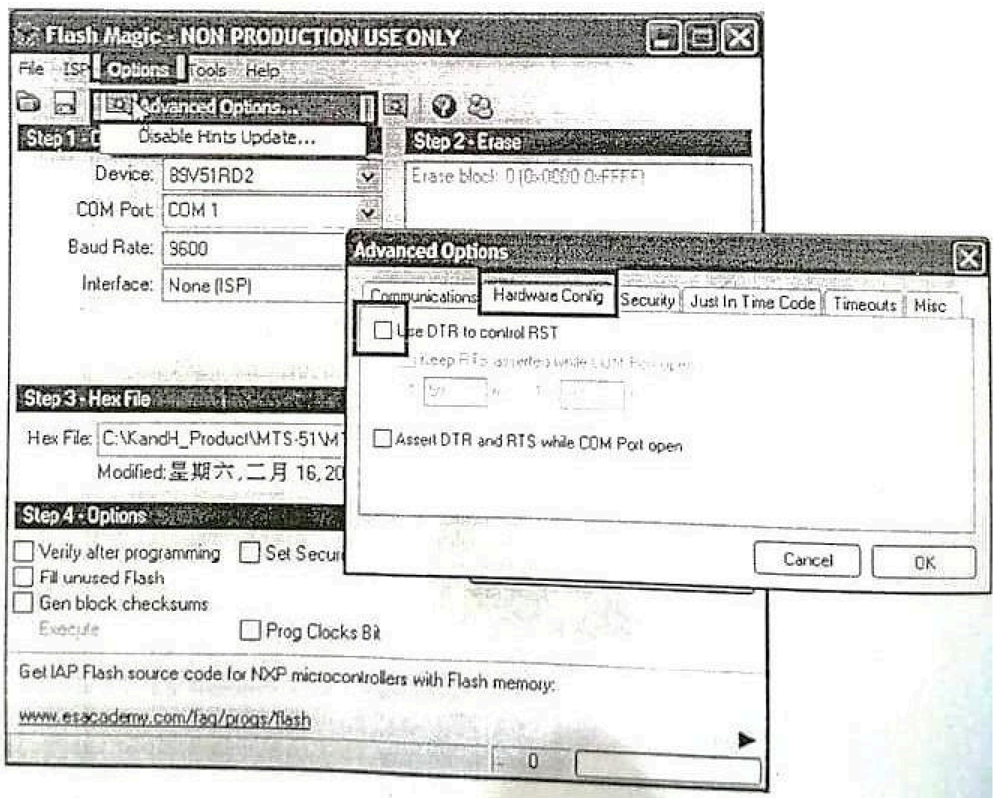
(3) Press Start Button



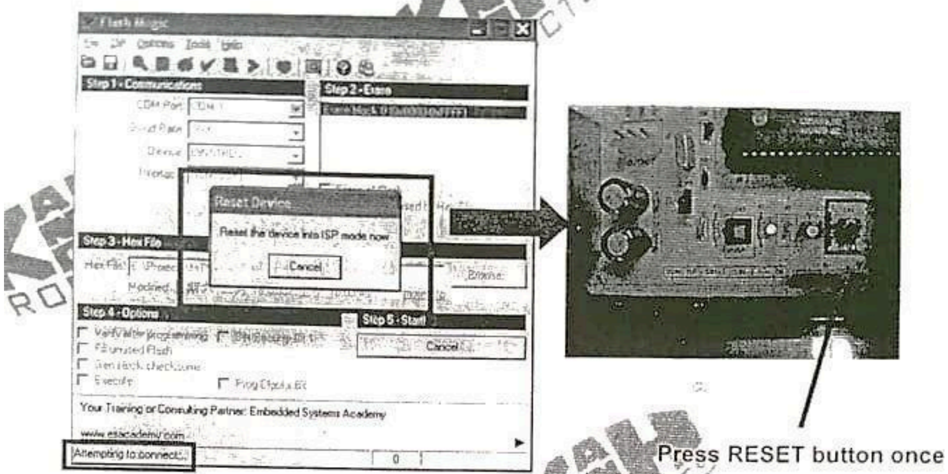
(Note: When you first time use Flash Magic 5.15 Version, you may see this window after pressed Start button.)



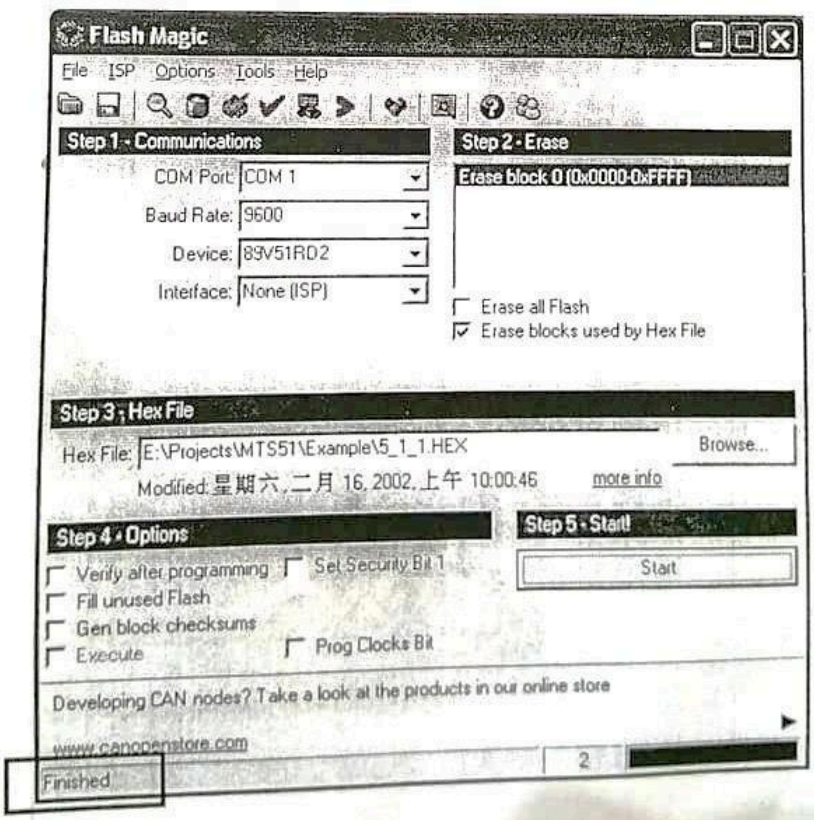
**(If you see above window, please go to Options => Advanced Options, and choose Hardware Config tab. Uncheck "Use DTR to control RST, and press OK.) (Press "Start" button again to download a program)



(4) After pressing Start Button, the Reset Device Window will pop up. At this time, please press RESET button on MTS-51 once.



(5) Once complete the download, the "Finished" message will show at bottom-left corner.



3. **Examine the Result**

   On MTS-51 Trainer, release SW5, place SW1 in correct position and then press the RESET button. Examine the result of program execution.

4. **Checks If Programming Failed**

   (1) Is the chip type selected correct?
   (2) Is the COM port correct?
   (3) Is the SW1 in correct position? （RS-232 TXD3 and RXD3）
   (4) Does RS232 driver/receiver chip MAX232 operate normally? Using VOM, measure the voltages on MAX232 pins 2 and 6, the voltage values on pins 2 and 6 should be +10V and -10V, respectively.

Lab Tasks:

1. Show the output of demo example 1 for LED DISPLAY CONTROL by configuring switches as you done in previous lab. Write down the steps to have a successful run. Take the snapshot of the final output generated by example.
2. Show the output of demo example 2 for 7 SEGMENT DISPLAY CONTROL (DS3 AND DS4) by configuring switches as you done in previous lab. Write down the steps to have a successful run. Take the snapshot of the final output generated by example.
3. Show the output of demo example 3 for DOT MATRIX LED CONTROL by configuring switches as you done in previous lab. Write down the steps to have a successful run. Take the snapshot of the final output generated by example.
4. Show the output of demo example 4 for MATRIX KEYBOARD CONTROL on DS4 by configuring switches as you done in previous lab. Write down the steps to have a successful run. Take the snapshot of the final output generated by example.
5. Show the output of demo example 5 for STEP MOTOR CONTROL by configuring switches as you done in previous lab. Write down the steps to have a successful run. Take the snapshot of the final output generated by example.
6. Show the output of demo example 6 for PHOTO INTERRUPTER CONTROL (JP4=PH2->T0) on STEP MOTOR by configuring switches as you done in previous lab. Write down the steps to have a successful run. Take the snapshot of the final output generated by example.

7. Show the output of demo example 7 for PULSE COUNTER (JP4=T0->OSC) on DS3 AND DS4 by configuring switches as you done in previous lab. Write down the steps to have a successful run. Take the snapshot of the final output generated by example.
8. Show the output of demo example 8 for TIMER/COUNTER CONTROL on LED BAR by configuring switches as you done in previous lab. Write down the steps to have a successful run. Take the snapshot of the final output generated by example.
9. Show the output of demo example 9 for SPEAKER CONTROL by configuring switches as you done in previous lab. Write down the steps to have a successful run. Take the snapshot of the final output generated by example.
10. Show the output of demo example 10 for LCM DISPLAY CONTROL by configuring switches as you done in previous lab. Write down the steps to have a successful run. Take the snapshot of the final output generated by example.

Lab Performance Evaluation:
Read and practice the manual. Performance is evaluated at the end of lab based on successfully running and understanding of examples, tasks and viva using defined rubrics.

Lab Report Evaluation:
Submit the Lab Report by writing all the examples and tasks which must include the following:

1. Brief description (3-5 lines)
2. The steps
3. The results (Screenshot)

Important note: Lab Report must be according to the Lab Report Format available on Google Classroom and must be submitted on time. Two similar reports will get zero marks. So, don’t try to copy your fellow reports. Lab report must be submitted in group of three maximum.